

Notes: Megginson & Weiss (JF, 1991)

Venture Capitalist Certification in Initial Public Offerings

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1 Big picture

- **Question.** Can venture capitalists certify the value of IPO firms and thereby reduce the costs of going public?
- **Core mechanism.** IPO firms are informationally opaque. Insiders know more than outside investors and have incentives to overstate value. A credible third party with reputation and money at stake can reduce this asymmetry.
- **Why venture capitalists are candidates for certification.** Venture capitalists repeatedly bring firms to market, interact with underwriters, auditors, institutional investors, and entrepreneurs, and have reputational capital that can be damaged by false certification.
- **Why certification is credible.** Venture capitalists also retain large equity stakes before and after the IPO. Because they do not usually cash out completely, they bear the future cost of overpricing or misrepresenting the firm.
- **Empirical design.** The paper compares 320 VC-backed IPOs to 320 non-VC-backed IPOs, matched as closely as possible by industry and offering size, during 1983–1987.
- **Main findings.** VC-backed IPOs have lower underpricing, lower gross spreads, higher quality underwriters and auditors, higher institutional holdings, and substantial post-IPO VC retention.
- **Key implication.** Venture capitalists reduce both indirect IPO costs, through lower underpricing, and direct costs, through lower underwriter compensation.
- **Big conclusion.** Venture capital backing is not merely financing before the IPO. It is also a market-certification technology that helps young firms access public capital markets earlier and at lower total cost.

2 Incremental contribution of the paper

- **Extends the certification literature beyond underwriters and auditors.** Earlier certification work emphasized investment banks, auditors, or other intermediaries. This paper shows that venture capitalists can also certify IPO quality.
- **Adds a dual bonding logic.** The paper emphasizes both reputational bonding and financial bonding. VCs care about future deal flow and repeat access to the IPO market, and they also keep equity stakes that expose them to future losses from false certification.
- **Links private and public capital markets.** The study shows that private-market monitors and financiers can affect public-market issuance costs when portfolio companies go public.
- **Documents certification using multiple margins.** Certification is not inferred from one outcome only. The paper studies underwriter quality, auditor quality, institutional holdings, initial returns, gross spreads, and post-IPO VC ownership.
- **Introduces a matched-pair empirical strategy for VC certification.** Matching VC-backed and non-VC-backed IPOs by industry and offering size helps compare firms that face similar offering markets.
- **Shows that certification affects both investors and financial specialists.** Lower underpricing suggests certification to investors; lower gross spreads suggest certification to underwriters, reducing their due-diligence and liability costs.
- **Provides early evidence that VCs do not simply use IPOs to cash out.** Table IX shows that VCs retain significant post-offering positions, strengthening the interpretation that their certification is credible.
- **Creates a foundation for later venture-capital value-added research.** The paper is an early empirical anchor for viewing VC as governance, monitoring, certification, and network access, not just capital provision.

3 Data and measurement

3.1 Sample construction

- The starting universe is firm commitment IPOs from January 1983 through September 1987 in the Investment Dealer's Digest Corporate Database.
- The authors exclude financial institutions, savings and loans, reverse LBOs, and firms whose first-day trading price is unavailable.
- VC-backed firms are identified using the *Venture Capital Journal*.
- The final matched sample contains 320 VC-backed IPOs and 320 non-VC-backed IPOs.
- The matching is done as closely as possible by industry and offering size.

Interpretation: The sample is designed to isolate VC backing from industry and offering-size effects. This is important because IPO underpricing and underwriting costs vary substantially across industries and issue sizes.

3.2 Core variables

- **VC-backed indicator.** Equals one if the IPO firm has venture-capital backing.
- **Initial return.** First trading day closing or bid price minus offer price, divided by the offer price.
- **Gross spread.** Underwriter compensation as a percentage of the offering price.
- **Offering expenses.** Auditor, legal, printing, registration, and other miscellaneous costs as a percentage of the offer price.
- **Underwriter quality.** Market share of the lead underwriter in the IPO market.
- **Auditor quality.** Indicator for whether the IPO uses a Big Eight auditor.
- **Institutional following.** Institutional holdings and the number of institutional managers holding the IPO after the offering.
- **Firm age.** Years from incorporation date to offer date.
- **VC holdings.** VC ownership before and after the IPO, and VC sales at the IPO.

Interpretation: The variables map directly into the certification hypothesis: if VC backing certifies firm value, investors should demand less underpricing, underwriters should require lower compensation, and financial specialists should be more willing to associate with the issue.

3.3 Three testable hypotheses

1. VC-backed IPOs should attract higher-quality underwriters, higher-quality auditors, and more institutional investors.
2. VC-backed IPOs should have lower underpricing and lower direct offering costs.
3. Venture capitalists should retain significant holdings after the IPO, providing a bonding mechanism that makes certification credible.

Interpretation: The paper's evidence is persuasive because the hypotheses are complementary. Reputation explains why VCs can certify; retention explains why they cannot easily profit from false certification; lower costs show that market participants respond to certification.

4 Models and empirical specifications

4.1 Certification logic

The certification problem begins with asymmetric information between corporate insiders and public investors. Insiders may benefit from selling overpriced securities, so outside investors discount the issue unless a credible certifier verifies quality.

A certifier is credible when three conditions are satisfied:

1. The certifier has reputational capital at stake.
2. The value of reputational capital exceeds the one-time gain from false certification.
3. The certifier's services are costly and more valuable when information asymmetry is larger.

Interpretation: Venture capitalists plausibly satisfy all three conditions. They repeatedly bring firms public, they need future entrepreneur and investor trust, and their financing is costly and selective.

4.2 Difference-in-means tests

The paper first compares VC-backed and non-VC-backed IPOs using differences in means and van der Waerden normal scores tests:

$$\Delta Y = \bar{Y}_{VC} - \bar{Y}_{NonVC}.$$

Interpretation: These comparisons establish the basic patterns: VC-backed firms have different characteristics and lower IPO costs. The matched sample makes these comparisons more informative than an unmatched comparison.

4.3 Initial-return regression

The main regression for underpricing is:

$$InitialReturn_i = \alpha + \beta_1 TYPE_i + \beta_2 LOGAMT_i + \beta_3 MKTSHR_i + \beta_4 AGE_i + \varepsilon_i.$$

- $TYPE_i = 1$ if the IPO is VC-backed.
- $LOGAMT_i$ is the log of the amount offered.
- $MKTSHR_i$ is the lead underwriter's average market share.
- AGE_i is the age of the firm.

Interpretation: The certification hypothesis predicts $\beta_1 < 0$: holding size, underwriter quality, and firm age constant, VC-backed IPOs should require less first-day compensation to investors.

4.4 Gross-spread regression

The main regression for direct underwriting costs is:

$$GrossSpread_i = \alpha + \gamma_1 TYPE_i + \gamma_2 LOGAMT_i + \gamma_3 MKTSHR_i + \gamma_4 AGE_i + u_i.$$

Interpretation: The certification hypothesis predicts $\gamma_1 < 0$. If VCs reduce underwriter information-acquisition costs and liability risk, underwriters should accept lower compensation.

4.5 Ownership-retention test

The post-IPO ownership test is not a regression; it is a bonding diagnostic. The paper documents how much equity venture capitalists own before the offering, whether they sell at the offering, and how much they retain afterward.

Interpretation: Retention is crucial because it shows that VCs are not merely certifying and immediately exiting. The retained stake makes false certification costly.

5 Identification and empirical design

5.1 Core identification problem

The core challenge is selection: VC-backed companies are not randomly assigned. They may differ in quality, technology, growth opportunities, and information environments. The paper addresses this problem by matching and by testing several distinct implications of the certification channel.

5.2 Matched-pair design

- Each VC-backed IPO is matched with a non-VC-backed IPO in a similar industry and with similar offering size.
- Industry matching controls for technology, growth, and market conditions.
- Offering-size matching controls for economies of scale in underwriting and issuance costs.

Interpretation: The matched-pair design does not make the study randomized, but it reduces two major confounds: industry composition and issue-size differences.

5.3 Mechanism evidence

- Table IV shows repeated VC-underwriter relationships and board participation, supporting reputational and relational capital.
- Table V shows that VC-backed firms attract more reputable underwriters and auditors and more institutional investors.
- Table VI shows lower initial returns and lower offering costs.
- Tables VII and VIII show that VC backing remains associated with lower initial returns and lower gross spreads after controls.
- Table IX shows that VCs retain substantial holdings after the IPO, supporting credible bonding.

Interpretation: The evidence is strongest because it is internally consistent. It is not just that VC-backed IPOs have lower underpricing; the supporting institutional patterns also fit the certification mechanism.

5.4 What the design cannot fully prove

- VC backing may proxy for unobserved firm quality.
- Higher-quality companies may be selected into venture financing before the IPO.

- Venture capitalists provide monitoring, governance, strategic advice, and network access, not certification alone.
- The sample is from the 1983–1987 IPO market and may not fully generalize to later IPO waves.

Interpretation: The paper’s strongest claim is not that VC backing is randomly assigned. Its strongest claim is that the evidence is consistent with and strongly supportive of VC certification, especially because multiple predictions hold simultaneously.

6 Section-by-section study guide

- **Introduction.** Frames third-party certification as a solution to IPO information asymmetry.
- **Section I.** Explains why venture capitalists satisfy the conditions for credible certification.
- **Section II.** Describes the matched sample and shows that VC-backed IPOs are concentrated in technology-intensive industries.
- **Section III.** Tests whether VC-backed IPOs attract higher-quality underwriters, auditors, and institutional investors.
- **Section IV.** Tests whether VC-backed IPOs have lower underpricing and lower underwriting costs.
- **Section V.** Documents VC holdings before and after IPOs to test whether VCs bond their certification by retaining ownership.
- **Conclusion.** Summarizes evidence that VCs lower the total cost of going public.

7 Tables and figures

7.1 Table I: SIC Classification For Venture Capital and Non-Venture Capital Backed IPOs

Read:

- The matched sample contains 640 IPOs in total: 320 VC-backed and 320 non-VC-backed.
- SIC 357, office/computing/accounting machines, accounts for 154 IPOs or 24.1% of the sample.
- SIC 737, computer and data processing services, accounts for 70 IPOs or 10.9%.
- Miscellaneous business services, including biotech and pharmaceutical engineering, accounts for 52 IPOs or 8.1%.
- The “Other” category accounts for 184 IPOs or 28.7%.

Detailed interpretation: The sample is heavily tilted toward technology, computing, communications, medical devices, and other information-intensive sectors. These are exactly the kinds of firms for which third-party certification should be valuable, because public investors have difficulty evaluating young firms with intangible assets and uncertain growth opportunities.

Economic message: The table establishes the high-information-asymmetry setting. VC certification is most valuable where firm quality is hard to assess.

Table I
SIC Classification For Venture Capital and Non-Venture Capital Backed IPOs

SIC classification and percentage of the total sample in each industry for the matched sample of 320 VC backed and 320 non-VC backed IPOs issued from January 1983 through September 1987 as identified from Investment Dealer's Digest Corporate Database and the *Venture Capital Journal*.

SIC Code	Classification	Number of IPOs	Percentage of IPOs
283	Drugs	30	4.7%
357	Office, Computing & Accounting Machines	154	24.1%
366	Communication Equipment	30	4.7%
367	Electronic Components & Accessories	48	7.5%
382	Measuring & Controlling Instruments	12	1.9%
384	Surgical, Medical & Dental Instruments & Supplies	26	4.0%
581	Eating and Drinking Places	14	2.2%
599	Retail Stores Not Elsewhere Classified	10	1.6%
737	Computer and Data Processing Services	70	10.9%
739	Miscellaneous Business Services (Biotech and Pharmaceutical Engineering)	52	8.1%
808	Outpatient Care Facilities	10	1.6%
	Other	184	28.7%
	TOTAL	640	100.0%

Table II
Number of VC Backed and Non-VC Backed IPOs By Year

Year	Venture Capital Backed	Non-Venture Capital Backed
1983	104 (32.5%)	137 (42.8%)
1984	47 (14.7%)	42 (13.1%)
1985	36 (11.2%)	44 (13.8%)
1986	78 (24.4%)	58 (18.1%)
1987	55 (17.2%)	39 (12.2%)
TOTAL	320	320

7.2 Table II: Number of VC Backed and Non-VC Backed IPOs By Year

Read:

- The sample is balanced by construction: 320 VC-backed and 320 non-VC-backed IPOs.
- 1983 is the largest year in the sample: 104 VC-backed IPOs and 137 non-VC-backed IPOs.
- The remaining years cover 1984–1987 with a mix of VC-backed and non-VC-backed offerings.
- The sample covers a five-year IPO cycle rather than a single hot-issue episode.

Detailed interpretation: The year distribution matters because underpricing can vary with IPO-market conditions. By showing sample coverage across several years, the table helps readers assess whether the results are likely to be driven by one narrow calendar period.

Economic message: The sample is time-balanced enough to compare VC-backed and non-VC-backed IPOs across a broader 1980s IPO market.

7.3 Table III: Tests of Differences in Sample Descriptive Statistics for VC Backed and Non-VC Backed IPOs

Read:

- VC-backed IPOs have a larger average amount offered: \$19.7 million versus \$13.2 million.
- VC-backed IPOs have a higher average offer price: \$11.18 versus \$10.16.
- Average preceding-year revenues are similar: \$37.1 million for VC-backed firms and \$39.4 million for non-VC-backed firms.
- Book asset means are not significantly different, but the median asset value is larger for VC-backed firms.
- VC-backed firms have higher common equity/assets: 41.7% versus 28.1%.
- VC-backed firms are younger: 8.6 years versus 12.2 years from incorporation to offer date.

Detailed interpretation: The table shows that the matched design does not eliminate all differences. VC-backed firms are younger and have a stronger equity base, which is consistent with growth companies going public earlier because VCs reduce information asymmetry. The age difference is especially important because young firms usually face more severe information problems.

Economic message: VC backing appears to help younger firms reach the IPO market earlier, while still obtaining larger offers and higher offer prices.

Table III
Tests of Differences in Sample Descriptive Statistics for VC Backed and Non-VC Backed IPOs^a

Tests of differences in offering characteristics using a difference in means test and a van der Waerden normal scores test for the sample of 320 VC backed and 320 non-VC backed IPOs matched as closely as possible by industry and offering size. Source: Investment Dealer's Digest Corporate Database and the offering prospectus.

Variable	Venture Capital Backed	Non-Venture Capital Backed	Difference in Means t-stat	van der Waerden Z score
Amount offered	\$19.7m [15.0m]	\$13.2m [9.2m]	5.20*	6.38*
Offering price	\$11.18 [10.50]	\$10.16 [10.00]	2.83*	3.41*
Preceding year's revenue	\$37.1m [16.2m]	\$39.4m [13.0m]	-0.33	1.49
Book value of assets	\$23.9m [12.9m]	\$27.2m [7.6m]	-0.76	3.90*
Growth in EPS per year	76.8% [61.1%]	65.5% [42.1%]	1.28	0.98
Total debt as a percentage of the book value of assets	31.3% [16.0%]	31.9% [21.5%]	-0.11	-2.61*
Book value of common equity as a percentage of the book value of assets	41.7% [44.8%]	28.1% [34.2%]	3.02*	3.70*
Years from incorporation date to offer date	8.6 yrs [5.3]	12.2 yrs [8.1]	-3.70*	-2.30**

^aMedians in brackets.

*Significant at the 0.01 level.

**Significant at the 0.05 level.

7.4 Table IV: Frequency of Board Participation, Percentage of Issues That the Venture Capitalist is the Lead, and Underwriter Selection for Venture Capitalists Who Brought 8 or More Issues to Market

Read:

- The most active venture capitalists repeatedly bring portfolio companies public.
- Kleiner Perkins brings 22 issues to market; Hambrecht & Quist Venture Partners brings 21.
- Many VCs are frequently on the board of directors of IPO firms, often in 50–100% of their issues.
- Many VCs are also lead investors in a substantial fraction of issues.
- The same underwriters appear repeatedly with the same venture capitalists.

Detailed interpretation: This table provides mechanism evidence. Certification requires repeated interaction and reputational capital. Frequent IPO involvement and repeated underwriter relationships make it costly for a VC to certify a low-quality firm falsely. Board participation and lead-investor status also suggest that the VC has real information about the firm, not just a passive financial stake.

Economic message: VC certification is credible because VCs are repeat players with relationships, reputations, information, and governance involvement.

Table IV
Board Participation, Percentage of Issues That the Venture Capitalist is the Lead or Selection for Venture Capitalists Who Brought 8 or More Issues to Market
Market from 1983 to 1987 for the venture capitalist, percentage of those issues for which the venture capitalist was underwriters used by the venture capitalist.

	Number of Issues Brought to Market	Percentage of issues the VC is on the Board of Directors	Percentage of issues the VC is the lead ^a	Most Frequent Underwriters ^b
ers	22	50%	27%	Robertson, Colman (9) Morgan Stanley (7)
tners	21	67%	38%	Hambrecht & Quist (14) Alex. Brown (4)
	15	40%	40%	Robertson, Colman (9)
	15	80%	33%	Robertson, Colman (6)
	14	0%	7%	Alex. Brown (5)
	14	86%	21%	Robertson, Colman (5) Morgan Stanley (6)
	13	77%	23%	Hambrecht & Quist (8) Morgan Stanley (5)
re	13	39%	0%	Morgan Stanley (6) Robertson, Colman (5)

(a) Table IV, part 1

Table IV—Continued

	Number of Issues Brought to Market	Percentage of issues the VC is on the Board of Directors	Percentage of issues the VC is the lead ^a	Most Frequent Underwriters ^b
	10	80%	30%	Alex. Brown (5) Morgan Stanley (5)
	10	90%	20%	Alex. Brown (5) Robertson, Colman (5)
rp	9	22%	22%	Alex. Brown (3)
	9	44%	44%	Hambrecht & Quist (3) Robertson, Colman (3)
	9	67%	0%	Hambrecht & Quist (4) Alex. Brown (2)
	8	63%	38%	Alex. Brown (2)
	8	100%	62%	Alex. Brown (4)
	8	100%	50%	Robertson, Colman (4)

(b) Table IV, continued

7.5 Table V: Tests of Differences in Mean Institutional Holdings, Average Market Share of Lead Underwriters and Auditor Quality for VC Backed and Non-VC Backed IPOs

Read:

- VC-backed IPOs use lead underwriters with higher average market share: 4.4% versus 3.0%.
- Lead underwriters for VC-backed firms bring more IPOs to market on average: 62 versus 53.
- 83% of VC-backed firms use Big Eight auditors, compared with 64% of non-VC-backed firms.
- Institutional holdings are much higher for VC-backed IPOs: 42.3% versus 22.2%.
- The average number of institutional managers is 10.2 for VC-backed IPOs and 5.4 for non-VC-backed IPOs.

Detailed interpretation: The table supports the certification channel on three margins. Underwriters, auditors, and institutions are all specialists who can either rely on or amplify the VC's certification. The higher institutional participation is especially important because it shows that sophisticated investors find VC-backed issues more credible or easier to evaluate.

Economic message: VCs help IPO firms attract higher-quality financial intermediaries and more institutional investors, lowering search and information costs in the offering process.

Table V
Tests of Differences in Mean Institutional Holdings, Average Market Share of Lead Underwriters and Auditor Quality for VC Backed and Non-VC Backed IPOs^a
 Tests of differences in institutional holdings from *Spectrum 3: Institutional Holdings* at the end of the first quarter following the offer date, the lead underwriters' market share and auditor quality for the sample of 320 VC backed and 320 non-VC backed IPOs matched by industry and size using a difference in means test and a van der Waerden normal scores test. The market share of each lead underwriter is computed using the full 2,644 IPOs in the IDD sample from 1983 to 1987. The total dollar amount of IPOs that are brought to market over the time period for each lead underwriter is divided by the total dollar amount of all IPOs issued (\$70.3 billion). The market shares of all lead underwriters for a particular IPO are then averaged. The differences in the frequency of use of Big Eight auditors is tested using a chi-square test.

Variable	Venture Capital Backed	Non-Venture Capital Backed	Difference in Means <i>t</i> -stat	van der Waerden <i>Z</i> -score
Average market share of lead underwriters	4.4% [3.4%]	3.0% [0.8%]	4.30*	6.02*
Average number of IPOs brought to market by lead underwriter	62 [60]	53 [38]	2.56*	2.98*
Number of IPOs using a big eight auditor	267 (83%)	219 (64%)		Chi-Square 19.89*
Average institutional holdings as a percentage of the amount offered for the quarter-end following the offer date	42.3% [39.5%]	22.2% [14.3%]	9.02*	9.35*
Average number of institutional managers	10.2 [8.0]	5.4 [4.0]	8.09*	8.72*

^aPercentage of firms in parentheses. Medians in brackets.

*Significant at the 0.01 level.

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7.6 Table VI: Tests of Differences in Initial Returns and Offering Expenses for VC Backed and Non-VC Backed IPOs

Read:

- In the matched sample, mean initial return is 7.1% for VC-backed IPOs and 11.9% for non-VC-backed IPOs.
- The average gross spread is 7.4% for VC-backed IPOs and 8.2% for non-VC-backed IPOs.
- Miscellaneous offering expenses are also lower for VC-backed IPOs: 3.6% versus 4.3%.
- The ratio of net proceeds to offering amount is higher for VC-backed IPOs: 0.89 versus 0.87.
- In the broader comparison with all eligible non-VC IPOs, the initial-return difference is much smaller and insignificant, highlighting the importance of matched comparisons.

Detailed interpretation: This is the central cost-of-going-public evidence. Lower underpricing benefits the issuing firm by reducing the wealth transfer to initial investors. Lower gross spreads and offering expenses reduce direct issuance costs. Together, these results imply that VC certification increases the net proceeds retained by the issuer.

Economic message: VC backing lowers both the indirect cost of IPO underpricing and the direct cost of underwriting and offering expenses.

Table VI
Tests of Differences in Initial Returns and Offering Expenses for VC Backed and Non-VC Backed IPOs
 Tests of differences in initial return using a difference in means test and a van der Waerden normal scores test for (1) the sample of all (991) non-VC IPOs with offer prices $\geq \$5$ and amount offered $\geq \$3$ million and the 320 VC backed firms and (2) for the matched sample of 320 VC backed and 320 non-VC backed issues matched by industry and offering size. Also presented are differences in offering expenses and total proceeds (excluding overallotment). Initial returns are defined as the first trading day close or bid price recorded in *Standard and Poor's Daily Stock Price Record: Over the Counter* minus the offer price divided by the offer price.

	Venture Capital Backed	Non-Venture Capital Backed	Difference in Means t-stat	van der Waerden Z Score
Panel A: Comparison of the Matched Sample of VC Backed and Non-VC Backed IPOs				
Number of firms	320	320		
Mean initial return	7.1% [2.5%]	11.9% [3.6%]	-3.62*	-1.87**
Percentage of IPOs with positive initial returns	61.2%	60.3%		
Average gross spread as a percentage of the offer price	7.4% [7.1%]	8.2% [7.3%]	-7.82*	-6.14*
Average of miscellaneous offering expenses as a percentage of the offer price ^a	3.6% [3.0%]	4.3% [3.6%]	-3.50*	-3.11*
Ratio of net proceeds to the offering amount ^b	0.89 [0.90]	0.87 [0.88]	7.28*	6.44*
Ratio of net proceeds to the first trading day amount ^{b,c}	0.84 [0.86]	0.80 [0.83]	4.97*	4.02*
Panel B: Comparison of VC Backed IPOs and All Non-VC Backed Firms with Price $\geq \$5$ and Amount Offered $\geq \$3$ million				
Number of firms	320	991		
Mean initial return	7.1% [2.5%]	7.6% [1.6%]	-0.52	-0.86

^aMiscellaneous offering expenses include such expenditures as auditor, legal, printing, and registration fees.
^bNet proceeds is defined as the offering amount minus underwriter fees and miscellaneous offering expenses.
^cThe first trading day amount is calculated as the number of shares offered multiplied by the first trading day price.
 *Significant at the 0.01 level.

can be both positive and negative. Furthermore, Ritter (1991) has documented

Table VII
OLS Regressions of Initial Returns (R_1) against Whether or Not the Issue is VC Backed (TYPE), the Log of the Amount Offered (LOGAMT), the Average Market Share of the Lead Underwriters (MKTSHR), and the Age of the Firm (AGE) for the Matched Sample of 320 VC Backed and 320 Non-VC Backed IPOs during the Period 1983-1987^a

$$R_1 = \alpha_0 + \alpha_1 TYPE + \alpha_2 LOGAMT + \alpha_3 MKTSHR + \alpha_4 AGE + e_i^b$$

Regression	α_0	α_1	α_2	α_3	α_4	R ²	F-Statistic	Significance of F-Test
(1)	0.137 (5.81)*	-0.045 (-2.95)*	-0.008 (-0.84)			0.018	5.7	0.0036
(2)	0.116 (4.86)*	-0.412 (-2.76)*	0.012 (1.19)	-0.847 (-4.18)*		0.044	9.7	0.0001
(3)	0.141 (5.66)*	-0.052 (-3.37)*	0.011 (1.03)	-0.722 (-3.55)*	-0.002 (-3.00)*	0.059	9.4	0.0001

^at-Statistics are given in parentheses.
^b R_1 = percentage return from the offer price to the first trading day price,
 TYPE = dummy variable for whether the IPO is VC backed (VC backed = 1, non-VC backed = 0),
 LOGAMT = log of offering amount (excluding the overallotment option) in millions,
 MKTSHR = average market share of all lead underwriters for a particular IPO defined as the percentage of all IPOs brought to market by each underwriter from 1983-1987 for the full sample of 2,644 IPOs, and
 AGE = age of the firm in years from incorporation date to offer date.
 *Significant at the 0.01 level.

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7.7 Table VII: OLS Regressions of Initial Returns against Whether or Not the Issue is VC Backed and Controls

Read:

- The dependent variable is initial return.
- The key independent variable is whether the IPO is VC-backed.
- Regressions control for offering size, underwriter market share, and firm age.
- VC backing is negatively related to initial returns in the matched sample.
- Underwriter quality is also negatively related to initial returns.

Detailed interpretation: The regression controls are important because Table III shows that VC-backed firms are not identical to non-VC-backed firms. The negative coefficient on VC backing after controls strengthens the claim that VC backing is associated with lower underpricing rather than merely reflecting issue size or underwriter reputation.

Economic message: The certification effect survives multivariate controls: VC-backed IPOs require less first-day underpricing compensation.

Tests of the certification hypothesis as it applies to initial returns and gross spreads using a regression analysis are presented in Tables VII and VIII. We examine the relationship of both initial returns and gross spreads to the following variables:

- (1) *A dummy variable for whether or not the issue is VC backed (TYPE).* The level of initial returns should be lower for VC backed issues (TYPE = 1) than for non-VC backed offers (TYPE = 0). In addition, we hypothesize that the presence of venture capitalists certifies the issue and lowers the cost of due diligence. Therefore, there should be a negative relationship between the dummy variable and both initial returns and gross spreads.
- (2) *The natural log of the amount offered (LOGAMT).* Ritter (1984) has documented a significant relationship between the size of the offering and initial returns. Since VC backed firms tend to be larger in terms of offering amount, controlling for size allows us to measure the relative influence of VC backing. If the presence of venture capitalists lowers initial returns, then the coefficient on this variable should be insignificant and negative. Furthermore, Ritter (1987) has documented

Table VII
OLS Regressions of Initial Returns (R_1) against Whether or Not
the Issue is VC Backed (TYPE), the Log of the Amount Offered
(LOGAMT), the Average Market Share of the Lead
Underwriters (MKTSHR), and the Age of the Firm (AGE) for the
Matched Sample of 320 VC Backed and 320 Non-VC Backed
IPOs during the Period 1983–1987^a
 $R_1 = \alpha_0 + \alpha_1 TYPE + \alpha_2 LOGAMT + \alpha_3 MKTSHR + \alpha_4 AGE + e_i^b$

Regression	α_0	α_1	α_2	α_3	α_4	R^2	F -Statistic	Significance of F -Test
(1)	0.137 (5.81)*	-0.045 (-2.95)*	-0.008 (-0.84)			0.018	5.7	0.0036
(2)	0.116 (4.86)*	-0.412 (-2.76)*	0.012 (1.19)	-0.847 (-4.18)*		0.044	9.7	0.0001
(3)	0.141 (5.66)*	-0.052 (-3.37)*	0.011 (1.03)	-0.722 (-3.55)*	-0.002 (-3.00)*	0.059	9.4	0.0001

^a t -Statistics are given in parentheses.

^b R_1 = percentage return from the offer price to the first trading day price,

TYPE = dummy variable for whether the IPO is VC backed (VC backed = 1, non-VC backed = 0),

LOGAMT = log of offering amount (excluding the overallotment option) in millions,

MKTSHR = average market share of all lead underwriters for a particular IPO defined as

7.8 Table VIII: OLS Regressions of Gross Spreads against Whether or Not the Issue is VC Backed and Controls

Read:

- The dependent variable is gross spread as a percentage of the offer price.
- VC backing is negatively related to gross spreads.
- Larger offering size lowers the percentage gross spread, consistent with economies of scale.
- Higher underwriter market share lowers gross spreads.

- Older firms also tend to have lower gross spreads, consistent with lower information asymmetry.

Detailed interpretation: The table shows that VC certification affects underwriters as well as investors. If VCs reduce the underwriter's information acquisition cost and legal/reputational risk, the underwriter can charge a lower percentage spread.

Economic message: VC certification lowers the direct compensation paid to underwriters, not only the investor discount embedded in underpricing.

Table VIII
OLS Regressions of Gross Spreads (GRSPD) against Whether or Not the Issue is VC Backed (TYPE), the Log Of the Amount Offered (LOGAMT), the Average Market Share of the Lead Underwriters (MKTSHR), and the Age of the Firm (AGE) for the Matched Sample of 320 VC Backed and 320 Non-VC Backed IPOs during the Period 1983–1987^a
 $GRSPD = \alpha_0 + \alpha_1 TYPE + \alpha_2 LOGAMT + \alpha_3 MKTSHR + \alpha_4 AGE + e_i^b$

Regression	α_0	α_1	α_2	α_3	α_4	R ²	Significance F-Statistic	of F-Test
(1)	0.106 (109.53)*	-0.003 (-4.31)*	-0.011 (-28.48)*			0.600	474.9	0.0001
(2)	0.105 (108.94)*	-0.002 (-4.09)*	-0.010 (-22.99)*	-0.052 (-6.35)*		0.622	349.7	0.0001
(3)	0.106 (108.40)*	-0.003 (-5.39)*	-0.010 (-22.69)*	-0.045 (-5.60)*	-0.0001 (-6.20)*	0.649	277.4	0.0001

^at-Statistics are given in parentheses.
^bGRSPD = gross spread paid to the underwriter as a percentage of the offer price,
TYPE = dummy variable for whether the IPO is VC backed (VC backed = 1, non-VC backed = 0),
LOGAMT = log of offering amount (excluding the overallotment option) in millions,
MKTSHR = average market share of all lead underwriters for a particular IPO defined as the percentage of all IPOs brought to market by each underwriter from 1983–1987 for the full sample of 2,644 IPOs, and
AGE = age of the firm in years from incorporation date to offer date.
*Significant at the 0.01 level.

economies of scale in the costs of going public. If this is the case, underwriter compensation should be negatively related to the size of

Table IX
Pre- and Post-Offering Characteristics of VC Holdings^a

	All VC IPOs (N = 320)
Panel A: Pre-Offering Characteristics	
Average percentage of equity owned by all venture capitalists prior to the offer	36.6% [36.3%]
Number of IPOs where venture capitalists own 50% or more of the shares prior to the offer	89 (28%)
Panel B: Post-Offering Characteristics	
Number of firms which have venture capitalists selling at the IPO	139 (43.3%)
Number of venture capitalists who sell 100% of their holdings	3 (0.9%)
Average percentage of venture capitalists' holdings sold at the IPO	8.0% [0.0%]
Average percentage of the amount offered that is composed of venture capitalists' shares	6.9% [0.0%]
Average percentage of secondary shares sold by venture capitalists (only for the 181 VC backed IPOs with secondary shares)	39.6% [33.8%]
Average percentage of the firm held by venture capitalists after the offer	26.3% [25.9%]
Number of IPOs where venture capitalists own 50% or more of the shares after the offer	27 (8.4%)

^aMedians in brackets. Percentage of firms in parentheses.

7.9 Table IX: Pre- and Post-Offering Characteristics of VC Holdings

Read:

- Before the IPO, venture capitalists own on average 36.6% of the firm's equity.
- In 89 IPOs, or 28% of the sample, VCs own at least 50% before the offer.
- Only 43.3% of VC-backed IPOs have any VC selling at the offering.
- Only three IPOs have VCs selling 100% of their holdings.
- VCs sell only 8.0% of their holdings on average at the IPO.
- After the IPO, VCs still own on average 26.3% of the firm.

Detailed interpretation: This table is the bonding evidence. If VCs intended to exploit overpricing, they would be expected to sell aggressively at the IPO. Instead, they retain large stakes and remain exposed to post-IPO firm value. This makes their certification credible.

Economic message: VCs are not simply cashing out at the IPO. Their retained holdings provide financial bonding for the credibility of certification.

Table IX
Pre- and Post-Offering Characteristics of VC Holdings^a

	All VC IPOs (N = 320)
Panel A: Pre-Offering Characteristics	
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Number of IPOs where venture capitalists own 50% or more of the shares prior to the offer	89 (28%)
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Number of firms which have venture capitalists selling at the IPO	139 (43.3%)
Number of venture capitalists who sell 100% of their holdings	3 (0.9%)
Average percentage of venture capitalists' holdings sold at the IPO	8.0% [0.0%]
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Average percentage of secondary shares sold by venture capitalists (only for the 181 VC backed IPOs with secondary shares)	39.6% [33.8%]
Average percentage of the firm held by venture capitalists after the offer	26.3% [25.9%]
Number of IPOs where venture capitalists own 50% or more of the shares after the offer	27 (8.4%)

^aMedians in brackets. Percentage of firms in parentheses.

7.10 Appendix A: Comparison of the Top Underwriters by Dollar Amount Market Share to the Carter and Manaster Ranking

Read:

- The appendix compares underwriter market-share rankings to the Carter-Manaster prestige rankings.
- Major underwriters such as Merrill Lynch, Lehman Brothers, Salomon Brothers, Goldman Sachs, and First Boston have high market shares and high prestige rankings.
- The table validates the use of underwriter market share as a quality proxy.

Detailed interpretation: Because underwriter quality is central to the paper's mechanism and regressions, the appendix helps establish that the market-share proxy is aligned with an established prestige measure.

Economic message: The appendix supports the measurement of underwriter reputation, one of the paper's key controls and mechanism variables.

following than non-VC backed IPOs.

By reducing the asymmetry of information between the issuing firm and investors and financial specialists such as underwriters and auditors, venture capitalists are able to lower the costs of going public. We find evidence of significantly lower underpricing and underwriter compensation, holding offering size, underwriter quality, and firm age constant for VC backed IPOs than for non-VC backed firms. Evidence on subsequent ownership structure of VC backed IPOs indicates that venture capitalists are not using the IPO as an opportunity to cash out of their holdings and realize a return on investment. Indeed, a majority of venture capitalists do not sell *any* of their holdings at the offer date.

Appendix A
Comparison of the Top Underwriters by Dollar Amount
Market Share to the Carter and Manaster (1990) Ranking

Underwriter	Market Share ^a	Carter and Manaster Rankings
Merrill Lynch Capital Markets	22.3%	9.0
Lehman Brothers	17.9	8.0
Saloman Brothers	15.4	9.0
Goldman, Sachs and Co.	14.7	9.0
First Boston Corp.	13.9	9.0
E. F. Hutton	11.2	8.0
Morgan Stanley	9.4	9.0
Drexel Burnham Lambert	8.9	7.0
Prudential Bache Securities	7.9	8.0
Alex. Brown and Sons	7.4	7.5
Dean Witter Reynolds	6.8	8.0
Bear, Stearns and Co.	6.1	8.0
Kidder Peabody and Co., Inc.	5.6	8.0
Paine Webber	4.6	7.5
Lazard Freres and Co.	4.4	8.0
Smith Barney, Harris Upham and Co.	3.7	8.0

^a in %

^b 1-10

8 Result map

- **Certification setup.** VCs have reputational capital, repeat-market interactions, and retained equity stakes.
- **Sample composition.** Tables I–III show that VC-backed IPOs are concentrated in high-information-asymmetry industries and are younger than controls.
- **Relationship mechanism.** Table IV shows repeated VC-underwriter relationships and governance involvement.
- **Financial specialist mechanism.** Table V shows higher-quality underwriters, higher-quality auditors, and more institutional ownership.
- **Cost outcome.** Table VI shows lower underpricing, lower gross spreads, and higher net proceeds.
- **Regression evidence.** Tables VII and VIII show that VC backing remains associated with lower initial returns and gross spreads after controls.
- **Bonding evidence.** Table IX shows that VCs retain large post-IPO stakes and generally do not cash out.

9 Short summary

- The paper tests whether venture capitalists certify IPO quality.
- Certification matters because IPO issuers are informationally opaque and insiders may overstate firm value.
- VCs can certify because they are repeat players with reputational capital and future deal flow at stake.
- VCs also hold large equity stakes, so false certification can hurt them financially.

- The empirical sample contains 320 VC-backed IPOs and 320 non-VC-backed matched controls from 1983–1987.
- VC-backed firms are younger, suggesting that VC certification helps firms access public markets earlier.
- VC-backed firms attract more reputable underwriters and auditors.
- VC-backed firms have more institutional following.
- VC-backed IPOs have lower underpricing and lower gross spreads.
- Regression results support the same conclusion after controlling for size, underwriter quality, and age.
- VCs retain substantial ownership after the IPO, so they are not merely cashing out.
- The paper’s main contribution is to show that venture capitalists are credible certifiers in public equity issuance.

10 Conclusion

10.1 Core conclusion

Meggison and Weiss provide evidence that venture capitalists certify IPO firms. Their presence lowers both indirect and direct costs of going public: underpricing falls, underwriter compensation falls, and issuing firms retain more of the proceeds.

10.2 Economic intuition

Young IPO firms face severe asymmetric information. Public investors cannot easily distinguish high- and low-quality firms, and insiders have incentives to overprice. Venture capitalists reduce this problem because they repeatedly interact with the IPO market, depend on their reputation for future deal flow, and continue to hold significant equity stakes after the issue.

10.3 Incremental contribution for corporate finance

The paper shifts the view of venture capital from purely pre-IPO financing to a broader certification and governance role. It shows that the value of VC backing is visible not only in firm growth or monitoring but also in the terms on which the firm enters public markets. This bridges venture capital, IPO underpricing, investment banking reputation, auditor choice, and institutional investor demand.

10.4 Policy and market implications

The results suggest that intermediaries with reputation and financial exposure can improve capital-market access for opaque firms. In markets where public investors have limited information, credible certifiers can reduce issuance costs. The evidence also implies that disclosure regulation alone may not substitute for certification by repeat private-market specialists.

10.5 Limitations

The paper cannot fully eliminate selection into venture backing. VCs may choose inherently better firms, and VC-backed firms may differ in unobserved ways from controls. The evidence is from the 1980s IPO market, so the exact magnitude of the certification effect may differ in modern IPO markets with different underwriters, institutions, and disclosure technologies.

10.6 Takeaway

The central takeaway is that venture capitalists create value at the IPO not just by selecting or monitoring companies before the offering, but by certifying them to public investors and financial intermediaries. The combination of reputation, repeated relationships, and retained ownership makes VC certification credible.